National University of Computer and Emerging Sciences
Lahore Campus

Programming Fundamentals (CS1002)

Sessional-II

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Course Instructor(s)

Ms. Anosha Khan, Dr Aamir Wali,
Mr. Razi Uddin, Ms. Arooj Khalil,
Ms Namra Absar, Mr. Uzair Naqvi,
Ms. Hina Iqbal, Mr. Zeeshan Nazar

Total Time (Hrs): 1
Total Marks: 30
Total Questions: 2

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Instructions:

Indent your code. Solve questions in the given order.

CLO#3: Apply algorithmic solutions related to the degree program to recent related problems.

Q1. Convolution is an operation that combines two arrays to produce a third sequence. Suppose we have 2 arrays:

Input array I = [a_0, a_1, a_2, a_3, a_4]
Filter array F = [f_0, f_1, f_2]

To perform convolution:

1. Place/align the filter on top of the first three elements of the array, that is [a_0, a_1, a_2].
2. Multiply corresponding pair ($a_0 * f_0$, $a_1 * f_1$, $a_2 * f_2$) and **sum them up**.
3. Move the filter one step to the right such that it is placed on [a_1, a_2, a_3].
4. Multiply corresponding pair ($a_1 * f_0$, $a_2 * f_1$, $a_3 * f_2$) and **sum them up**.
5. Repeat until the filter no longer fully overlaps the array.
6.

Sample example
I = [1, 2, 3, 4, 5, 6]
F = [1, 4, 2]

Step	Elements Used (from I)	Calculation	Result
1	(1, 2, 3)	(1×1) + (2×4) + (3×2)	1 + 8 + 6 = **15**
2	(2, 3, 4)	(2×1) + (3×4) + (4×2)	2 + 12 + 8 = **22**
3	(3, 4, 5)	(3×1) + (4×4) + (5×2)	3 + 16 + 10 = **29**
4	(4, 5, 6)	(4×1) + (5×4) + (6×2)	4 + 20 + 12 = **36**

i=m
while (i
n=0
n+1

Output [15, 22, 29, 36]

int x=0
int y=c
i++)

Implement the function void convolution (int input[], int sizeInput, int filter[], int sizeFilter) that performs the **convolution** of a 1-D input array with a filter and prints the resulting convolved data. Assume sizeInput > sizeFilter. **You don't need to implement the main function**.

(14+1[indentation] = 15 marks)

Q2. Implement the function series (int x, int n) that takes in integers x and n as input parameter. The function **prints** the first n terms of the series and the **sum** of the n term:

(2+2+2+6[series]+3[main+indentation]) = 15 marks)

$$\frac{1!}{x^1} + \frac{3!}{x^3} + \frac{5!}{x^5} + \frac{7!}{x^7} + \cdots$$

For this problem, you need to come up with **three user-defined functions** (besides main and series) to perform subtasks of the computation. One of these functions must be **PrintTerm(...) that prints a term** of the series. To print all n terms, this function will be called n times. Each function must make logical sense and should handle a specific part of the logic. The function series (...) must call them appropriately.

✓input
✓print
fact
sum
main

Sample example

Input x = 2, n = 3
Output:

1! / 2 ^ 1+3! / 2 ^ 3+5! / 2 ^ 5+

Sum = 5

Note: Use ^ for power. The + at the end of the series is deliberate.

int numerator = factorial

n = 3+2+1